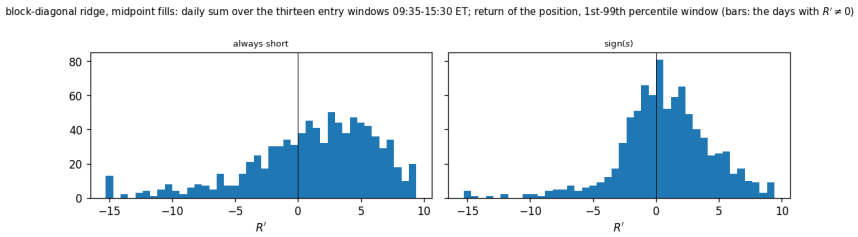
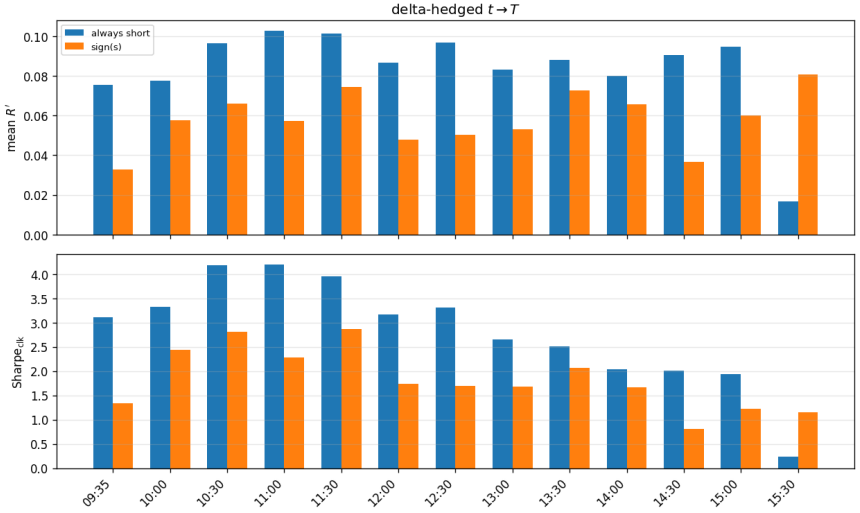
0DTE nearest-OTM straddle: per-rule return summary, pooled over the thirteen entry windows 09:35–15:30 ET (overlapping remaining-session books), models compared on the same 866 days

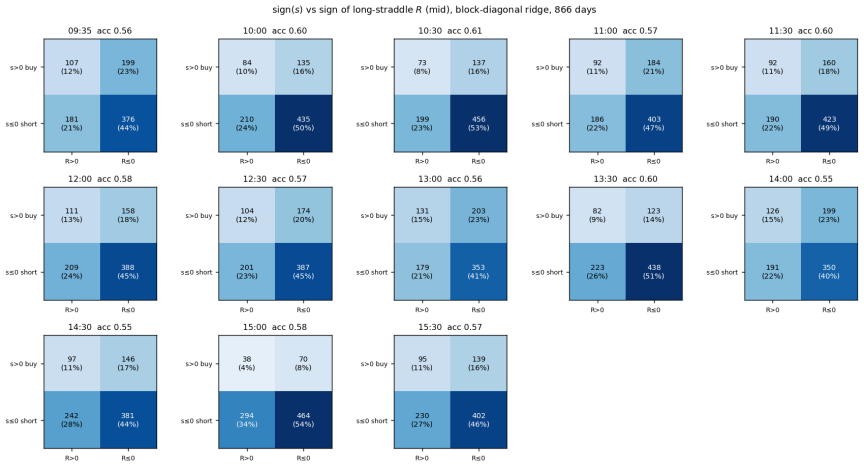
	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{ann}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{ann}}$
Short volatility (always short)												
all models (no forecast)	866	1.090	5.746	-58.68	10.56	-2.51	16.34	5.58	3.01	0	0.0	2.28
sign(s) *(± 1 on the sign of s)*												
baseline (HAR + calendar OLS)	866	0.660	3.896	-26.83	33.98	-0.17	9.67	4.99	2.69	143	16.5	1.61
block-diagonal ridge	866	0.755	4.053	-26.83	41.81	0.19	16.37	5.48	2.96	179	20.7	1.92
block-diagonal ridge, without the FOMC columns	866	0.755	4.022	-26.83	41.81	0.24	16.65	5.52	2.98	192	22.2	1.94
LightGBM†	866	0.752	3.936	-26.83	41.81	0.41	17.58	5.63	3.03	147	17.0	1.97
XGBoost†	866	0.764	3.729	-26.83	23.05	-0.77	6.81	6.03	3.25	148	17.1	2.13
lasso (causally tuned)†	866	0.783	4.152	-26.83	52.74	1.34	32.20	5.55	3.00	159	18.4	1.98
lasso ($\alpha = 10^{-4}$)	866	0.730	3.983	-26.83	33.92	-0.40	10.24	5.39	2.91	187	21.6	1.86
elastic net (causally tuned)†	866	0.728	4.214	-26.83	52.74	1.19	30.81	5.08	2.74	162	18.7	1.74



Distribution of the return of the daily sum over the thirteen entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and $\text{Sharpe}_{\text{clk}}$ by entry clock (grouped bars). $\text{Sharpe}_{\text{clk}}$ is that clock's daily series $\times\sqrt{252}$, not the pooled $\text{Sharpe}_{\text{ann}}$. Delta-hedged $t \to T$: hold entry K to official close, rebalance Δ every 30 minutes (from 09:35 the first rebalance is at 10:00).



Confusion matrix per entry clock: sign(s) (buy if $s > 0$) against the sign of the long-straddle midpoint return. 15:30 is the last-30-min trade.

Each row is the daily sum of the thirteen overlapping remaining-session books: enter at each clock 09:35–15:30 ET, hold those strikes to the official close, delta-hedge every 30 minutes (from 09:35 the first re-hedge is at 10:00). 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's thirteen overlapping remaining-session books. That reserved name is the book's daily P&L after pooling the session. The thirteen books overlap; they are not thirteen successive holds. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the thirteen positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat. At the 09:35 entry the forecast is the one issued at 09:30 for the 09:30–10:00 bar, w_t is that bar's share, and h_t is the chain's hours to the close.